

the Kalman or H_∞ performance measures can be given. Analytical determination of stability and performance bounds for this type of filter is an open research issue.

12.2 ROBUST KALMAN/H_∞ FILTERING

The material in this section is based on [Hun03]. In most practical problems, an exact model of the system may not be available. The performance of the system in the presence of model uncertainties becomes an important issue. For example, suppose we have a system given as

$$\begin{aligned} x_{k+1} &= (F_k + \Delta F_k)x_k + w_k \\ y_k &= (H_k + \Delta H_k)x_k + v_k \end{aligned} \quad (12.14)$$

where $\{w_k\}$ and $\{v_k\}$ are uncorrelated zero-mean white noise processes with covariances Q_k and R_k , respectively. Matrices ΔF_k and ΔH_k represent uncertainties in the system and measurement matrices. These uncertainties are assumed to be of the form

$$\begin{bmatrix} \Delta F_k \\ \Delta H_k \end{bmatrix} = \begin{bmatrix} M_{1k} \\ M_{2k} \end{bmatrix} \Gamma_k N_k \quad (12.15)$$

where M_{1k} , M_{2k} , and N_k are known matrices, and Γ_k is an unknown matrix satisfying the bound

$$\Gamma_k^T \Gamma_k \leq I \quad (12.16)$$

[Recall that we use the general notation $A \leq B$ to denote that $(A - B)$ is a negative semidefinite matrix.] Assume that F_k is nonsingular. This assumption is not too restrictive; F_k should always be nonsingular for a real system because it comes from the matrix exponential of the system matrix of a continuous-time system, and the matrix exponential is always nonsingular (see Sections 1.2 and 1.4). The problem is to design a state estimator of the form

$$\hat{x}_{k+1} = \hat{F}_k \hat{x}_k + K_k y_k \quad (12.17)$$

with the following characteristics:

1. The estimator is stable (i.e., the eigenvalues of $\hat{F}_k$ are less than one in magnitude).
2. The estimation error $\tilde{x}_k$ satisfies the following worst-case bound:

$$\max_{w_k, v_k} \frac{\|\tilde{x}_k\|_2}{\|w_k\|_2 + \|v_k\|_2 + \|\tilde{x}_0\|_{S_1^{-1}} + \|x_0\|_{S_2^{-1}}} < \frac{1}{\theta} \quad (12.18)$$

3. The estimation error $\tilde{x}_k$ satisfies the following RMS bound:

$$E(\tilde{x}_k \tilde{x}_k^T) < P_k \quad (12.19)$$

The solution to the problem can be found by the following procedure [Hun03].

The robust mixed Kalman/ H_∞ filter

1. Choose some scalar sequence $\alpha_k > 0$, and a small scalar $\epsilon > 0$.
2. Define the following matrices:

$$\begin{aligned} R_{11k} &= Q_k + \alpha_k M_{1k} M_{1k}^T \\ R_{12k} &= \alpha_k M_{1k} M_{2k}^T \\ R_{22k} &= R_k + \alpha_k M_{2k} M_{2k}^T \end{aligned} \quad (12.20)$$

3. Initialize P_k and $\tilde{P}_k$ as follows:

$$\begin{aligned} P_0 &= S_1 \\ \tilde{P}_0 &= S_2 \end{aligned} \quad (12.21)$$

4. Find positive definite solutions P_k and $\tilde{P}_k$ satisfying the following Riccati equations:

$$\begin{aligned} P_{k+1} &= F_{1k} T_k F_{1k}^T + R_{11k} + R_{11k} R_{2k} R_{11k}^T - \\ &\quad [F_{1k} T_k H_{1k}^T + R_{11k} R_{2k} R_{12k} + R_{12k}] R_k^{-1} [\cdot \cdot \cdot]^T + \epsilon I \\ \tilde{P}_{k+1} &= F_k \tilde{P}_k F_k^T + F_k \tilde{P}_k N_k^T (\alpha_k I - N_k \tilde{P}_k N_k^T)^{-1} N_k \tilde{P}_k F_k^T + \\ &\quad R_{11k} + \epsilon I \end{aligned} \quad (12.22)$$

where the matrices R_{1k} , R_{2k} , F_{1k} , H_{1k} , and T_k are defined as

$$\begin{aligned} R_{1k} &= (\tilde{P}_k^{-1} - N_k^T N_k / \alpha_k)^{-1} F_k^T \\ R_{2k} &= R_{1k}^{-1} (\tilde{P}_k^{-1} - N_k^T N_k / \alpha_k)^{-1} R_{1k}^{-T} \\ F_{1k} &= F_k + R_{11k} R_{1k}^{-1} \\ H_{1k} &= H_k + R_{12k}^T R_{1k}^{-1} \\ T_k &= (P_k^{-1} - \theta^2 I)^{-1} \end{aligned} \quad (12.23)$$

5. If the Riccati equation solutions satisfy

$$\begin{aligned} \frac{1}{\theta^2} I &> P_k \\ \alpha_k I &> N_k \tilde{P}_k N_k^T \end{aligned} \quad (12.24)$$

then the estimator of Equation (12.17) solves the problem with

$$\begin{aligned} K_k &= [F_{1k} T_k H_{1k}^T + R_{11k} R_{2k} R_{12k} + R_{12k}] \tilde{R}_k^{-1} \\ \tilde{R}_k &= H_{1k} T_k H_{1k}^T + R_{12k}^T R_{2k} R_{12k} + R_{22k} \\ \hat{F}_k &= F_{1k} - K_k H_{1k} \end{aligned} \quad (12.25)$$

The parameter ϵ is generally chosen as a very small positive number. In the example in [Hun03] the value is $\epsilon = 10^{-8}$. The parameter α_k has to be chosen large enough so that the conditions of Equation (12.24) are satisfied. However, as α_k increases, P_k also increases, which results in a looser bound on the RMS estimation error.

A steady-state robust filter can be obtained by letting $P_{k+1} = P_k$ and $\tilde{P}_{k+1} = \tilde{P}_k$ in Equation (12.22) and removing all the time subscripts (assuming that the system is time-invariant). But the resulting coupled steady-state Riccati equations will be more difficult to solve than the discrete-time Riccati equations in Equation (12.22), which can be solved by a simple (albeit tedious) iterative process. Similar problems have been solved in [Mah04b, Xie04, Yoo04].

■ EXAMPLE 12.2

Suppose we have an angular positioning system such as a motor. The moment of inertia of the motor and its load is J and the coefficient of viscous friction is B . The torque that is applied to the motor is $cu + w$, where u is the applied voltage, c is a motor constant that relates applied voltage to generated torque, and w is unmodeled torque that can be considered as noise. The differential equation for this system is given as

$$J\ddot{\phi} + B\dot{\phi} = cu + w \quad (12.26)$$

where ϕ is the motor shaft angle. We choose the states as $x(1) = \phi$ and $x(2) = \dot{\phi}$. The dynamic system model can then be written as

$$\begin{aligned} \dot{x} &= Ax + B_u u + B_w w \\ &= \begin{bmatrix} 0 & 1 \\ 0 & -B/J \end{bmatrix} x + \begin{bmatrix} 0 \\ c/J \end{bmatrix} u + \begin{bmatrix} 0 \\ 1/J \end{bmatrix} w \end{aligned} \quad (12.27)$$

In order to discretize the system with a sample time T , we use the method of Section 1.4 to obtain

$$x_{k+1} = Fx_k + G_u u_k + G_w w_k \quad (12.28)$$

The discrete-time system matrices are given as

$$\begin{aligned} F &= \exp(AT) \\ G_u &= \int_0^T \exp(At) dt B_u \\ &= \frac{c}{B} \begin{bmatrix} T - 1/\alpha + e^{-\alpha T}/\alpha \\ 1 - e^{-\alpha T} \end{bmatrix} \\ G_w &= \frac{1}{B} \begin{bmatrix} T - 1/\alpha + e^{-\alpha T}/\alpha \\ 1 - e^{-\alpha T} \end{bmatrix} \\ \alpha &= \frac{B}{J} \end{aligned} \quad (12.29)$$

If our measurement is angular position ϕ corrupted by noise, then our measurement equation can be written as

$$y_k = \begin{bmatrix} 1 & 0 \end{bmatrix} x_k + v_k \quad (12.30)$$

Suppose the system has a torque disturbance w_k with a standard deviation of 2, and a measurement noise v_k with a standard deviation of 0.2 degrees. We can run the Kalman filter and the robust mixed Kalman/H_∞ filter for

this problem. Figure 12.2 shows the position and velocity estimation errors of the Kalman and robust mixed Kalman/ H_∞ filters. The robust filter performs better at the beginning of the simulation, although the Kalman filter performs better in steady state. (It is not easy to see the advantage of Kalman filter during steady state in Figure 12.2 because of the scale, but over the last half of the plot the Kalman filter estimation errors have standard deviations of 0.33 deg and 1.65 deg/s, whereas the robust filter has standard deviations of 0.36 deg and 4.83 deg/s.)

Now suppose that the moment of inertia of the motor changes by a factor of 100. That is, the filter assumes that J is 100 times greater than it really is. In this case Figure 12.3 shows the position and velocity estimation errors of the Kalman and robust filters. It is apparent that in this case the robust filter performs better not only at the beginning of the simulation, but in steady state also. After the filters reach “steady state” (which we have defined somewhat arbitrarily as the time at which the position estimation error magnitude falls below 1 degree) the Kalman filter RMS estimation errors are 0.36 degrees for position and 1.33 degrees/s for velocity, whereas the robust filter RMS estimation errors are 0.28 degrees for position and 1.29 degrees/s for velocity. The square roots of the diagonal elements of the P_k Riccati equation solution of Equation (12.22) reach steady-state values of 0.51 degrees and 1.52 degrees/s, which shows that the estimation-error variance is indeed bounded by the Riccati equation solution P_k .

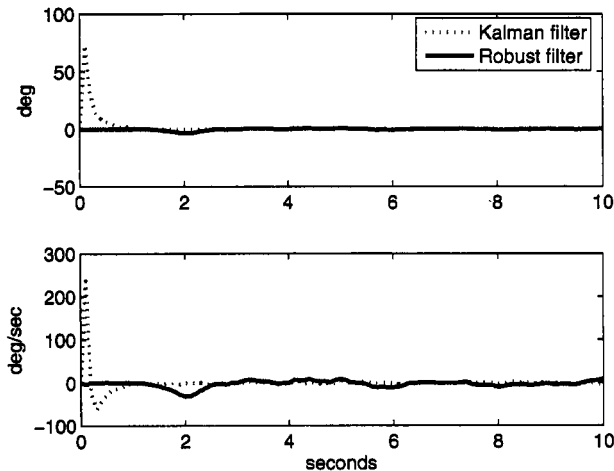


Figure 12.2 Position and velocity estimation errors for Example 12.2 for the Kalman filter and the robust filter, assuming that the system model is perfectly known. The robust filter performs better at the beginning of the simulation, but the Kalman filter performs better in steady state. The steady-state Kalman filter estimation errors have standard deviations of 0.33 deg and 1.65 deg/s, whereas the robust filter has standard deviations of 0.36 deg and 4.83 deg/s.